

(%i2) /* Assume h, p be positive (important for absolute values later) */

assume(h > 0)\$;

assume(p > 0)\$;

(%i5) /* Expand the solution in Taylor series, taking only the lead term. Choose h, h/2 and h/4 */

eq1: y = y_h + C · h^p; /* h step size */

eq2: y = y_h_2 + C · (h / 2)^p; /* h/2 step size */

eq3: y = y_h_4 + C · (h / 4)^p; /* h/4 step size */

eq1 $y = y_h + C h^p$

eq2 $y = y_{h_2} + \frac{C h^p}{2^p}$

eq3 $y = y_{h_4} + \frac{C h^p}{4^p}$

(%i8) /* Compute C twice */

C_1: factor(solve(rhs(eq1) = rhs(eq2), C)[1]);

factor(solve(rhs(eq2) = rhs(eq3), C)[1])\$;

C_2: factor(ratsubst(2^p · 2^p, 4^p, %)); /* Simplifying 4^p */

C_1 $C = \frac{2^p (y_{h_2} - y_h)}{(2^p - 1) h^p}$

C_2 $C = \frac{2^{2p} (y_{h_4} - y_{h_2})}{(2^p - 1) h^p}$

(%i10) /* Compute the absolute errors */

abs(subst([C = rhs(C_1)], eq1) - y_h); /* Error of y_h */

abs(subst([C = rhs(C_1)], eq2) - y_h_2); /* Error of y_h/2 */

(%o9) $|y_h - y| = \frac{2^p |y_{h_2} - y_h|}{2^p - 1}$

(%o10) $|y_{h_2} - y| = \frac{|y_{h_4} - y_{h_2}|}{2^p - 1}$

(%i12) /* C_1 must be equal to C_2 when asymptotic regime holds. We get the following condition */

abs((factor(rhs(C_1) = rhs(C_2))) · h^p / 2^p · (2^p - 1) / (y_h_4 - y_h_2))\$;

ratsubst(2^p · 2^p, 4^p, %); /* Simplifying 4^p */

(%o12) $\frac{|y_{h_2} - y_h|}{|y_{h_4} - y_{h_2}|} = 2^p$